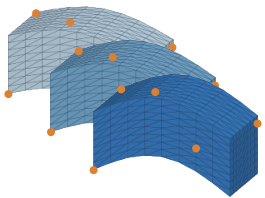
OBSERVE & FIT



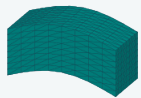
Position-track prefix

Geometry, mass, camera and tool poses supplied

Fit scene descriptors

COMPOSE PHYSICAL ROLES

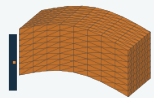
Material coordinates



$W(F, z_m)$

Bulk response

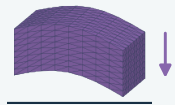
Tool-relative



$\psi(\delta, z_b)$

Attachments

World coordinates



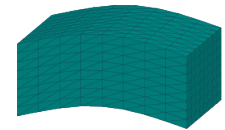
$\Pi_c + \Pi_g$

Contact + gravity

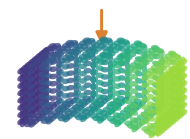
Mechanics: compose energies and solve for motion

FREEZE & QUERY

New tool positions



Positions / deformation



Gaussian surface

Illustrative schematic

Controlled simulations

Frozen bulk energy
Q2 mechanics + dynamic contact

Real PhysTwin captures

Neural / analytic equilibrium
Shared Gaussian transport

SoftVTBench manipulation

Analytic shape + correction
Slip / release + reduced contact

Shared physical roles; distinct benchmark adapters. Appearance is not refitted to future images.